MIL-R-23452A(AS)
19 July 1982
SUPERSEDING
MIL-R-23452(WEP)
7 November 1962

MILITARY SPECIFICATION

ROCKET MOTOR MARK 25 MODS 0, 1, AND 2

This specification is approved for use by the Naval Air Systems Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the Rocket Motor Mark 25 Mods 0, 1, and 2 (see 6.1).

1.2 Classification. The Rocket Motor Mark 25 will consist of the following Mods:

Mod 0 - 30° Canted nozzle
Mod 1 - 15° Canted nozzle
Mod 2 - 0° Canted nozzle

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-23453	Igniter, Rocket Motor, Mk 188 Mod 0
MIL-G-23454	Grain, Propellant, Rocket Motor, Mk 66 Mod 0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department, (ESSD) Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization/ Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

2185549	Palletized Shipping Container, Rocket Motor Mk 25 Mods
2202762	Identification Marking
2202766	Rocket Motor Mk 25 Mod 0 Major Assembly
2202768	Chamber
2205571	Rocket Motor Mk 25 Mod 1 Major Assembly
2205572	Rocket Motor Mk 25 Mod 2 Major Assembly
LD 537843	Rocket Motor Mk 25 Mod 0
LD 537844	Rocket Motor Mk 25 Mod 1
LD 537845	Rocket Motor Mk 25 Mod 2

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.1.1 First article samples. Unless otherwise specified in the contract, the contractor shall manufacture a lot of 30 rocket motors, using the same materials, processes, and equipment proposed for production. Twenty-eight rocket motors are to be used for the tests scheduled in Section 4, and two motors are to be spares. The first article samples shall meet the requirements of Section 3 when tested in accordance with Section 4.

3.2 Materials. Materials shall be in accordance with the applicable drawings and specifications as specified on List of Drawings as follows:

Mark 25 Mod 0 - LD 537843
Mark 25 Mod 1 - LD 537844
Mark 25 Mod 2 - LD 537845

Substitution of materials shall not be made without prior consent of the procuring activity. The use of substitute materials shall not, however, constitute a waiver of any requirement or inspection provision of this specification.

3.2.1 Propellant. The propellant grain, Mark 66 Mod 0, shall be in accordance with MIL-G-23454.

3.2.2 Igniter. The igniter, Mark 188 Mod 0, shall be in accordance with MIL-I-23453.

3.2.3 Attachment fittings. The ultimate strength of the attachment fittings shall meet the requirements of Drawing 2202768 covering a detail of Drawings 2202766, 2205571, and 2205572.

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3.3 Performance characteristics.

3.3.1 Ballistic test. The performance of the unit shall comply with applicable ballistic test requirements shown in Table I.

3.3.2 Thermal cycling. The unit shall meet the requirements of 3.3.1 when tested in accordance with 4.4.2.2.2 or 4.4.2.3.2.

3.3.3 Accelerated aging. The unit shall meet the requirements of 3.3.1 when tested in accordance with 4.4.3.

3.3.4 Vibration. The unit shall meet the requirements of 3.3.1 when tested in accordance with 4.4.2.1, 4.4.2.2.1, and 4.4.2.3.1.

TABLE I. Ballistic test requirements.

Characteristics	Temperature, deg F		
	-40	60	150
Average Thrust, lb			
Min	2960	3740	4680
Max	4000	5060	6340
Maximum Thrust, lb	4300	5400	6900
Action Time, sec			
Min	5.62	4.55	3.71
Max	7.60	6.16	5.03
Total Impulse, lb-sec			
Min	21,670	22,220	22,690
Max	24,330	24,960	25,490
Ignition Interval, sec			
Max	0.325	0.200	0.200
Average Pressure, psig 1/			
Min	930		1440
Max	1270		1960
Maximum Peak Pressure, psig 1/	1325		2060

1/ Acceleration tests only.

3.3.5 Shock Tests. The unit shall meet the requirements of 3.3.1 when tested in accordance with 4.4.2.2.3 or 4.4.2.3.3.

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3.3.6 Acceleration. The unit shall meet the requirements of 3.3.1 when tested in accordance with 4.4.4.

3.3.7 Exhaust characteristics. The unit shall not eject fragments of propellant or other solid particles harmful to aircraft or turbojet engines, during or after rocket firing. The exhaust shall exhibit minimum toxic and corrosive characteristics.

3.3.8 Operating temperature. The operating temperature range shall be between -40 and 150°F.

3.4 Chemical and physical properties.

3.4.1 Envelope dimensions. Envelope dimensions and attachment points shall comply with Drawings 2202766, 2205571, or 2205572 as applicable.

3.4.2 Weight. The weight of the complete unit shall be no greater than 210 pounds.

3.5 Identification. Each unit produced under a single contract or purchase order shall be serially numbered. Records shall be maintained such that the batch or lot of propellant used in a motor can be identified by the serial number of the motor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Production acceptance tests. Acceptance tests shall be performed on each production lot of rocket motors.

4.2.1 Lots. For acceptance tests, a lot of rocket motors shall consist of units manufactured by a uniform process in which a homogeneous product would be expected. The maximum lot size may be specified at the option of the procuring activity (see 6.2). A lot of grains shall consist of all grains manufactured from one mix batch of propellant.

4.2.2 Inspection. The contractor shall, during the process of manufacture and assembly, perform such inspections and tests as required to assure conformance of component parts, subassemblies, and assemblies with

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applicable contract, order, requisition, drawing, or specification requirement. Characteristics classified on the drawings or in specifications as critical shall be inspected 100 percent. Unless otherwise specified in the contract, sampling for other characteristics shall be in accordance with one of the following:

a. MIL-STD-1235, CSP-1
b. MIL-STD-105
c. MIL-STD-414

The Acceptable Quality Level (AQL) shall be 1.0 percent for characteristics classified major and 6.5 percent for characteristics classified as minor.

4.2.3 Ballistic acceptance. Each lot of rocket motors shall be tested for ballistic performance by random selection and static firings of motors in accordance wtih 4.3 for Lot Acceptance Tests (LATs). Sample size shall be as specified in Figure I. One-third of the LATs shall be fired at 40°, 60°, and 150°F. The ballistic sample for reduced plan shall be fired alternately at -40°, 60°, and 150°F. Temperature conditioning of LATs shall be for not less than 48 hours. Each motor shall be fired within 20 minutes after removal from the conditioning chamber. Anytime the conditioning chamber temperature is not maintained within the specified limits or the motor is out of conditioning more than 20 minutes prior to firing, the motor shall be reconditioned at the specified temperature for a period twice the elapsed time, but not exceeding the required 48 hours. Acceptance or rejection of motor lots shall be based on compliance with the requirements for average thrust, maximum thrust, action time, total impulse, and ignition interval specified in Table I.

4.3 Ballistic firing test method. Rocket motors shall be fired on a thrust stand designed to permit accurate measurement of the required parameters. The thrust shall be continuously recorded from initial application of pulse until final tail-off. Resulting thrust-time curves must be such that data as required on the specific test may be derived. Required data for acceptance and first article tests are those listed in Table I.

4.4 First article tests. A preproduction lot of 28 rocket motors shall be subjected to tests in the quantities indicated and in accordance with the procedures stated herein. Igniters shall be installed in all motors except Groups III and IV during all preproduction testing. For Groups III and IV, the igniters shall be installed after completion of vibration testing. (The igniter boss is used to attach the text fixture through which the vibrational force is applied.) Each unit shall be radiographic inspected in accordance with MIL-STD-453 prior to subjection to preproduction testing. Each unit shall be weighed and the weight recorded before and after firing.

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4.4.1 Statistical tests. Eight motors shall be selected at random from the preproduction lot and divided into two groups of four motors each, designated Group I and Group II. Group I shall be conditioned to -40°F for 48 hours minimum and fired in accordance with 4.3 within 20 minutes after removal from the conditioning temperature. Group II shall be conditioned to 150°F for 48 hours minimum and fired in accordance with 4.3 within 20 minutes after removal from the conditioning temperature.

4.4.2 Sequential tests. Eight motors shall be selected at random from the preproduction lot and divided into two groups of four motors each, designated Group III and Group IV. Each motor shall be radiographic inspected in accordance with MIL-STD-453 prior to initiation of sequential tests. These motors shall be subjected to the environmental testing specified in the following paragraphs.

4.4.2.1 Resonant vibration survey. Each unit shall be subjected to a resonant frequency survey in the longitudinal direction. The survey shall be conducted by applying a 1 g input and slowly varying the frequency from 5 to 500 cps. Resonant frequencies, if existent, shall be noted. The surveys shall be conducted at -40° and 150°F. Resonant frequencies shall be defined as that point on the F/A vs frequency plot where the F/A values exceed a ratio of 1.5 at a clearly defined peak.

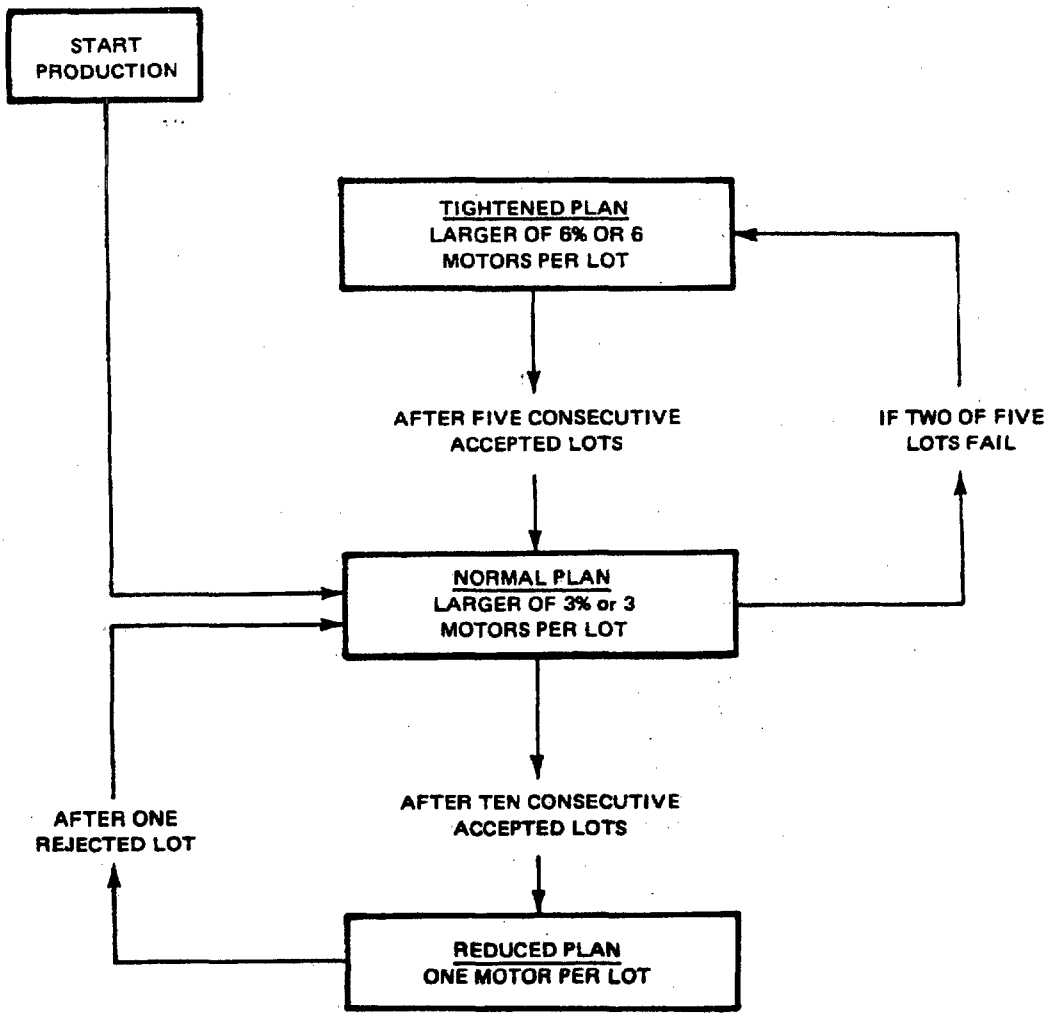
4.4.2.2 Group III.

4.4.2.2.1 Endurance vibration. Each unit in Group III shall be conditioned to -40°F and vibrated at -40°F with a nominal frequency of 30 cps and a nominal acceleration of 1 g for 24 hours. Each unit shall then be conditioned at 150°F and vibrated at 150°F at the most significant resonant frequency as determined in 4.4.2.1 for 24 hours at a nominal acceleration of 5 g. After completion of endurance vibration, each motor in Group III shall be radiographic inspected in accordance wtih MIL-STD-453.

4.4.2.2.2 Temperature cycling. Each unit in Group III shall be temperature cycled from -40° to 150°F for three complete cycles. The motors shall be maintained at each successive temperature for a minimum of 48 hours. The time of transfer between temperatures shall not exceed 10 minutes. After completion of temperature cycling, each motor in Group III shall be radiographic inspected in accordance with MIL-STD-453.

4.4.2.2.3 Shock. Each unit shall be conditioned to -40°F for 48 hours minimum. Each unit shall then be subjected to accelerations of 12 g normal to the axis and 30 g along the axis. Each acceleration shall consist of one-half sine wave form with a duration of 15 ± 5 milliseconds. After completion of shock tests, each motor in Group III shall be radiographic inspected in accordance with MIL-STD-453.

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NOTE: The term "Accepted Lot" shall apply to a lot that meets all requirements of Table I and does not apply to lots which fail one or more ballistic parameters but are subsequently accepted on a waiver.

FIGURE 1. Ballistic test sampling plan.

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4.4.2.2.4 Temperature gradient firing. Each unit shall be conditioned in a chamber controlled at -40°F for 48 hours minimum, then transferred to a chamber controlled at 150°F for 2 hours. The unit shall then be fired in accordance with 4.3 within 20 minutes from the time of removal from the temperature conditioning chamber.

4.4.2.3 Group IV.

4.4.2.3.1 Endurance vibration. Each unit in Group IV shall be conditioned at 105°F and vibrated at 150°F with a nominal frequency of 30 cps and a nominal acceleration of 1 g for 24 hours. Each unit shall then be conditioned at -40°F and vibrated at -40°F at the most significant resonant frequency as determined in 4.4.2.1 for 24 hours at a nominal acceleration of 5 g. After completion of endurance vibration each motor in Group IV shall be radiographic inspected in accordance with MIL-STD-453.

4.4.2.3.2 Temperature cycling. Each unit in Group III shall be temperature cycled from 150° to -40°F for three complete cycles. The motors shall be maintained in each successive temperature environment for a minimum of 48 hours. After completion of temperature cycling, each motor in Group IV shall be radiographic inspected in accordance with MIL-STD-453. The time of transfer between temperature conditioning rooms shall not exceed 10 minutes.

4.4.2.3.3 Shock. Each unit shall be conditioned to 150°F for 48 hours minimum. Each unit shall then be subjected to accelerations of 12 g normal to the axis and 30 g along the axis. Each acceleration shall consist of one-half sine wave form with a duration of 15 ± 5 milliseconds. After completion of shock tests, each motor in Group IV shall be radiographic inspected in accordance with MIL-STD-453.

4.4.2.3.4 Temperature gradient firing. Each unit shall be conditioned in a chamber controlled at 150°F for 48 hours minimum, then transferred to a chamber controlled at -40°F for 2 hours. The unit shall then be fired in accordance with 4.3 within 20 minutes from the time of removal from the temperature conditioning chamber.

4.4.3 Accelerated aging tests. Eight units shall be selected at random from the preproduction lot and divided into two groups of four motors each, designated Group V and Group VI. The units in Group V shall be stored at -40°F for 30 days. Each unit shall then be radiographic inspected in accordance with MIL-STD-453. Each unit shall then be conditioned for a minimum of 48 hours at -40°F and fired in accordance with 4.3 within 20 minutes after removal from the conditioning environment. The units in Group VI shall be stored at 150°F for 30 days. Each unit shall then be radiographic inspected in accordance with MIL-STD-453. Each unit shall then be conditioned for a minimum of 48 hours at 150°F and fired in accordance with 4.3 within 20 minutes after removal from the conditioning temperature.

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4.4.4 Acceleration tests. Four units shall be selected at random from the preproduction lot and divided into two groups of two motors each, designated Group VII and Group VIII. The units in Group VII shall be conditioned for 48 hours minimum in a temperature environment of -40°F. Following temperature conditioning, each unit shall be placed on a dynamic facility (track, centrifuge, or catapult), accelerated to 10 g along the longitudinal axis of the unit, and fired. Pressure-time data for parameters specified in Table I shall be obtained. The units in Group VIII shall be conditioned for 48 hours minimum in a temperature environment of 150°F. Following temperature conditioning, each unit shall be placed on a dynamic facility (track, centrifuge, or catapult), accelerated to 10 G along the longitudinal axis of the unit, and fired. Pressure-time data for parameters specified in Table I shall be obtained.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.1.1.1 Unit packaging. Unit packaging of the rocket motors shall be in accordance with LD 537844 and Drawing 2185549. The unit pallet shall contain nine rocket motors.

5.2 Packing.

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.2.1.1 Exterior containers. Nine rocket motors packaged as specified in 5.1.1.1 shall be packed in accordance with packaging drawings of LD 537844.

5.3 Marking.

5.3.1 Special markings. Marking of exterior containers shall be in accordance wtih the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with Drawing 2202762 and MIL-STD-129.

5.4 Storage. Rocket motors shall be capable of storage in any position for a period of 3 years in a storage temperature range of -40° to 140°F without deterioration.

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6. NOTES

6.1 Intended use. The solid rocket propellant motor covered by this specification is intended for jet assisted takeoff use on piloted aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Full name of item including Mark and Mod numbers
c. Preproduction lot size, when required
d. Maximum lot size, when required
e. Drawing numbers and other applicable documents
f. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.3 Definition of terms.

6.3.1 Ignition interval time. The ignition interval time shall be defined as the time interval from the application of current to the 75 percent of maximum thrust on the ascending portion of the thrust-time curve.

6.3.2 Maximum thrust. Maximum thrust shall be defined as the highest thrust recorded on the thrust-time trace.

6.3.3 Action time. Action time shall be defined as the time interval on the thrust-time record (pressure for acceleration firings) from 10 percent of maximum thrust (pressure) on the initial rise of the record to the corresponding 10 percent of maximum thrust (pressure) on the final declining portion of the record.

6.3.4 Total impulse. The total impulse shall be defined as the area under the thrust-time curve between the limits of the action time.

6.3.5 Average thrust. The average thrust shall be defined as the action time total impulse (see 6.3.4) divided by the action time (see 6.3.3).

6.3.6 Average pressure. Average pressure shall be defined as the value obtained by dividing pressure-time integral (see 6.3.7) by action time (see 6.3.3).

6.3.7 Pressure time integral. Pressure-time integral shall be defined as the integral of the pressure-time record between the limits of action time (see 6.3.3).

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6.4 First article. When a first article is required, the item will be sted and should be a first article sample. The first article should nsist of 30 units. The contracting officer should include specific structions in acquisition documents regarding arrangements for examition, test, and approval of the first article.

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